

Changyue (William) Zhao

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EDUCATION

Duke University, Durham, NC

08/2025 – Present

Master of Biostatistics

- GPA: 4.0/4.0
- Selected Coursework: *Statistical Methods for Learning and Discovery, Deep Learning for Health Data, High-Dimensional Statistics and Machine Learning, Real Analysis, Statistical Inference (Ph.D. Level)*
- Ranked second in Real Analysis by course grade (Instructor: Michael C. Reed)

University of Minnesota, Twin Cities, Minneapolis, MN

09/2023 – 05/2025

Bachelor of Science in Statistical Science, Minor in Computer Science

- GPA: 4.0/4.0
- Selected Coursework: *Mathematical Analysis, Theory of Statistics, Regression and Correlated Data, Statistical Machine Learning, Applied Multivariate Methods, Algorithms and Data Structures, Graduate Data Mining*
- Graduated with High Distinction and received the Buehler Memorial Undergraduate Award as one of eight nominees. Named to the CLA Dean's List from Fall 2023 to Spring 2025

Shandong University of Finance and Economics, Jinan, China

09/2021 – 06/2025

Bachelor of Economics in Finance

- GPA: 92.86/100 (WES GPA: 3.89/4.0)
- Selected Coursework: *Advanced Mathematics, Linear Algebra, Probability Theory and Statistics, Applied Statistics, Python, Microeconomics, Finance*
- Received the SDUFE First-Prize Scholarship in 2022 and 2023

RESEARCH INTERESTS

- Applications of foundation models and LLM-based agents for biomedical analysis
- Deep learning methods for biomedical data analysis and multimodal integration

MANUSCRIPT & THESIS

Changyue Zhao and Zhicheng Ji.

09/2025 – Present

Visual LLM-guided consensus spatial domain detection with L-STAR.

First-author manuscript accepted in principle at Nature Communications

- Developed the L-STAR visual LLM workflow for spatial domain detection, with structured prompts, repeated comparisons of spatial maps and optional histology, JSON logs, and win-rate-based method selection.
- Implemented consensus clustering with co-assignment matrices and average linkage. Added an all-wise comparison mode that aggregates repeated rankings to select ensemble members with one LLM call per repeat.
- Designed comparative experiments across six spatial transcriptomics datasets to examine ensemble size, alternative LLMs, and fixed-method consensus baselines, separating the roles of method selection and clustering aggregation.
- Built marker-gene-based domain annotation with spot- and cell-level labels and an Unknown option. Evaluated agreement with manual annotations using BioLORD-2023 embeddings and within-dataset permutation tests.
- Audited sensitivity to color palettes, presentation order, and redundant candidate methods by examining ranking agreement, selected-method overlap, and clustering stability through repeated runs and controlled perturbations.
- **Open-source software:** [L-STAR](#)  **Supervisor:** Prof. Zhicheng Ji, Duke University

Comparative Analysis of Forecasting Abilities of ARIMA, LSTM, and Transformer Models in the Financial Domain: A Case Study of the Stock Market and Cryptocurrency Market

03/2024 – 05/2025

- Collected and harmonized stock-market and cryptocurrency data from Wind and Yahoo Finance. Implemented ARIMA models in R and LSTM and Transformer models in PyTorch.
- Adapted Transformer input embeddings and feature weighting for financial time-series forecasting. Compared statistical and neural models across the Shanghai Composite, Nasdaq, and CMC200 indices.
- Received the Outstanding Graduation Thesis Award.
- **Supervisor:** Prof. Mengmeng Zheng, Shandong University of Finance and Economics

RESEARCH EXPERIENCE

LLM-Guided Pseudotime Analysis Agent

01/2026 – Present

Manuscript in preparation

- Developed **pseudotime-agent**, an end-to-end Python package using **LangGraph** to coordinate R/Seurat analysis and LLM inference, from evidence generation to per-cell pseudotime estimation through a configurable command-line interface.
- Designed parallel UMAP and t-SNE branches for cell-type annotation, concurrent lineage-family adjudication, and cross-embedding trajectory fusion, using targeted marker-gene and graph evidence to review uncertain lineage relationships.
- Extended the agent with a spatial pseudotime branch that combines tissue adjacency and molecular evidence through LLM-guided trajectory proposal, relation adjudication, ordering review, and comparison with nonspatial trajectories.
- Implemented expression-weighted shortest-path pseudotime for spatially separate instances of a shared process. Built synthetic negative controls and perturbation tests to assess spatial dependence and ordering stability.
- Implemented JSON schema validation, corrective LLM retries, resumable execution, and provenance logging. Added support for expert-supplied annotations and preprocessed Seurat inputs, including scATAC-seq gene-activity data.
- Designed and implemented **bPOS**, an exact branch-aware pseudotime metric that jointly evaluates lineage membership and within-lineage ordering using scalable block processing and inclusion-exclusion counting.
- Diagnosed a bug in the **dyneval**-based benchmark evaluation of **F1branches**. Reimplemented the published Jaccard, recovery, relevance, and harmonic-mean definition over the full cell scope and added diagnostic comparisons.
- Built reproducible evaluation workflows for comparisons with Monocle 3, Slingshot, and TSCAN and curated gold-standard, silver-standard, and synthetic trajectory datasets as benchmark resources.

Med2Vec+

11/2025 – Present

School of Medicine, Duke University, Durham, NC

- Developed Med2Vec+, a note-aware, multi-aspect extension of **Med2Vec** [🔗](#) that represents patient histories as time-by-aspect-by-code tensors and incorporates demographic covariates.
- Encoded clinical notes with **BioClinicalBERT** and aligned visit-level note embeddings with structured EHR events.
- Designed a multi-head cross-attention module that conditions structured EHR embeddings on clinical notes for context-aware patient representation learning.
- Implemented the **PyTorch** model with Med2Vec-compatible inputs and outputs and tested components through ablation.
- **Software:** [GitHub](#) [🔗](#), [Hugging Face](#) [🔗](#) **Supervisor:** Prof. Chuan Hong, Duke University

PaperEval

03/2025 – 09/2025

School of Statistics, University of Minnesota, Twin Cities, Minneapolis, MN

- Built **Selenium** and asynchronous data-collection pipelines for major machine-learning conferences on OpenReview, storing papers, reviews, scores, author responses, and metadata in **JSON** with resumable checkpoints.
- Extracted author contact information from conference PDFs with **PyMuPDF** to enrich OpenReview metadata.
- Fine-tuned GPT-4o for acceptance prediction using progressively richer inputs: paper metadata, reviewer scores, and review comments. Implemented training-data preparation and classification evaluation workflows.
- Developed LLM-based agent workflows for manuscript and figure assessment, review-response scoring, and response generation. Evaluated generated responses against author replies from accepted and rejected submissions.
- Designed a bounded Gaussian-noise mapping to convert acceptance levels into continuous constructiveness targets.
- **Open-source software:** [PaperEval](#) [🔗](#) **Supervisor:** Prof. Jie Ding, University of Minnesota

MUDAC (Midwest Undergraduate Data Analysis Competition) 2025

03/2025 – 04/2025

School of Statistics, University of Minnesota, Twin Cities, Minneapolis, MN

- Encoded sparse, high-dimensional numerical and text data with embeddings, sentiment scores, and pairwise similarities.
- Applied Cox proportional hazards regression and a causal Bayesian network to study match duration and closure.
- Designed a mixture-of-experts model with regression, MLP, and Transformer experts for different feature types and a learned gating network. Compared it with random forest and XGBoost baselines.
- Placed 7th in Innovation & Creativity.
- **Supervisors:** Barbara Kuzmak and Robert Oliver VandenBurg, University of Minnesota

University of Minnesota & Hennepin County Public Health Equity Data Challenge

10/2024 – 11/2024

School of Public Health, University of Minnesota, Twin Cities, Minneapolis, MN

- Analyzed Hennepin County questionnaire data on inequities in access to mental-health resources using multiple regression, ridge regression, decision trees, and causal inference methods.
- Built a mental-health prediction interface in **tkinter** backed by an **AdaBoost** model in **scikit-learn**.
- Co-authored the analysis report and poster and placed in the top 25% of the undergraduate division.
- **Supervisors:** Marta Shore and Prof. Julian Wolfson, University of Minnesota

Overlapping Group LASSO Regression R Package Implementation

09/2024 – 05/2025

School of Statistics, University of Minnesota, Twin Cities, Minneapolis, MN

- Compared `overlappingLeastR` from the MATLAB SLEP library with `oglasso` in the R package `VennLasso` under matched problem sizes and convergence tolerances.
- Simulated 1,000 large sparse-matrix datasets with overlapping groups and compared runtime using t-tests.
- Implemented the optimized algorithm in C for integration with R.
- **Supervisor:** Prof. Tianxi Li, University of Minnesota

Data-Intensive Research Projects

07/2024 – 06/2025

School of Economics and Management, Tsinghua University, Beijing, China

- Implemented a LinkedIn crawler for 7,063 profiles with `DrissionPage`, depth-first search, multithreading, persistent SQLite storage, and resumable checkpoints.
- Implemented error logging and automated gap recovery for a Google Patents collection pipeline handling approximately 300 GB of Chinese patent text.
- Conducted literature screening and synthesis on the effects of generative AI, reviewing 50 papers and organizing 13 core studies into a focused literature review.
- **Supervisor:** Prof. Wu Zhu, Tsinghua University

Study on Pricing of Carbon Emission Rights Trading in China

03/2023 – 06/2023

School of Finance, Shandong University of Finance and Economics, Jinan, Shandong, China

- Collected data from Wind and constructed macroeconomic, fossil-fuel price, and policy variables.
- Diagnosed underfitting in R regression models and examined associations between carbon prices and policy variables.
- Implemented vector autoregression and impulse-response analysis in R and LSTM forecasting in `TensorFlow`.
- **Supervisor:** Prof. Zumian Xiao, Shandong University of Finance and Economics

PROFESSIONAL EXPERIENCE

Teaching Assistant: BIOSAT 830

08/2026 – Present

Department of Biostatistics and Bioinformatics, Duke University, Durham, NC

Supervisor: Dr. Lin (Lynn) Lin

- Hold weekly office hours and grade assignments in graduate statistical inference, asymptotic theory, and decision theory.
- Prepare and verify proof-based LaTeX assignment solutions. Format and proofread lecture slides, standardizing mathematical notation and converting formulas into editable PowerPoint equations.

Undergraduate Teaching Assistant: STAT 3011

09/2024 – 05/2025

School of Statistics, University of Minnesota, Twin Cities, Minneapolis, MN

- Taught weekly 50-minute sections for about 30 students on probability, sampling, hypothesis testing, regression, and R.
- Developed instructional slides, held office hours, and assessed weekly assignments and exam responses through Gradescope. Shared teaching materials with the course team.
- Teaching-language qualification: ELP 1, the highest rating on the Spoken English Test for Teaching Assistants (equivalent to TOEFL iBT Speaking 27–30/30 or IELTS Speaking 8.5–9.0/9).
- **Materials:** [GitHub](#)  **Supervisors:** Yuyoung Park, Aritra Banerjee, Kazeem Adepoju, University of Minnesota

Intern, Management Department of Exclusive Agency (EA)

06/2024 – 08/2024

Huatai Property & Casualty Insurance Co., Ltd (A Chubb Company), Beijing, China

- Modeled customer value using decision trees and regression on one million exclusive-agency customer records, examining predictors to support customer segmentation.
- Co-developed a classifier for new customers and segmented existing customers using quadrant-based rules and k-means in R, matching the resulting segments to agencies.
- **Supervisor:** Lifeng Wang, Manager of Strategy and Innovation Development, Huatai P&C

OTHER INFORMATION

- **Programming:** Python, R, C/C++, Java, OCaml, Swift, MATLAB, SQL
- **Machine Learning:** PyTorch, TensorFlow, scikit-learn, Hugging Face
- **Scientific Computing:** Scanpy, Seurat, NumPy, pandas
- **Tools:** Git, SQLite, Selenium, LaTeX, Markdown, SPSS
- **Hobbies:** Photography, Snowboarding, Car Racing, Cycling, Powerlifting